

RAG CORPUS: RURAL FIRST AID PROTOCOLS IN MOROCCO##

1. General Principles and Moroccan Operational Framework

- * Moroccan Chain of Survival: According to the Ministry of Health of the Kingdom of Morocco, any first-aid intervention must follow the regulatory triad: Protect, Alert, Resuscitate. In rural Morocco, geographic isolation dictates that the first bystander must act immediately to stabilize vital functions before medical teams arrive.
- * Official Emergency Numbers in Morocco:
 - * 141: National toll-free number for the SAMU (Services d'Assistance Médicale Urgente), essential for medical regulation and dispatch.
 - * 15: Civil Protection (Fire and Rescue Services), responsible for victim rescue, extrication, and emergency medical evacuations in rural areas.
 - * 177: Royal Gendarmerie, the law enforcement authority to contact for motor vehicle accidents on rural roads.
- * Scene Safety and Secondary Accidents: The rescuer must first assess environment risks (fire, landslides, electrical hazards) to protect themselves, the victim, and bystanders. Securing the scene is a priority to prevent a secondary accident.

2. Choking (Foreign Body Airway Obstruction)

- * Clinical Diagnosis (Signs): The victim clutches their throat, cannot speak, coughs weakly, or cannot breathe at all. The face may turn cyanotic (blue).
- * Adult Protocol (Moroccan Red Crescent / WHO Standards):
 1. Give 5 sharp back blows using the heel of your hand between the victim's shoulder blades. Lean the victim forward.
 2. If back blows fail, perform 5 abdominal thrusts (Heimlich Maneuver): stand behind the victim, wrap your arms around their waist, place a fist just above the navel, grasp it with your other hand, and pull sharply inward and upward.
 3. Alternate 5 back blows and 5 abdominal thrusts until the object is expelled or the victim becomes unconscious.
- * Infant Protocol (Under 1 Year Old): Place the infant face down along your forearm, supporting the head with your hand. Deliver 5 back blows with the heel of your hand. If unsuccessful, flip the infant

face up and apply 5 chest thrusts using two fingers in the center of the chest.

* Action for Unconsciousness: Call SAMU (141) or Civil Protection (15) immediately and begin Cardiopulmonary Resuscitation (CPR).

3. Burns

* Emergency Classification: In rural Morocco, burns from butane gas cylinder explosions or boiling liquids are frequent. Gravity criteria include the surface area (larger than half of the victim's palm), specific locations (face, hands, joints, respiratory tract), and vulnerable ages (infants, elderly).

* Immediate Management Protocol:

1. Cool immediately: Apply the WHO rule of cooling. Run cool, low-pressure tap water (15°C to 20°C) over the burn for at least 10 to 15 minutes to stop heat from damaging deeper tissues.
2. Clothing removal: Remove non-adherent clothing while cooling the burn. Never tear off clothing that sticks to a deep burn.
3. Protection: Cover the burn loosely with a clean cloth or clean plastic wrap to prevent infection. Never apply traditional local remedies (oil, flour, toothpaste), as they complicate surgical treatment.
4. Alert: Call 15 or 141 for severe burns.

4. Bleeding (Severe Hemorrhage)

* Life-Threatening Emergency: Uncontrolled bleeding can cause death within minutes due to hypovolemic shock.

* Direct Pressure Protocol:

1. Manual pressure: Apply firm, continuous pressure directly over the bleeding wound using a clean cloth or gloved hand (to prevent bloodborne pathogen transmission).
2. Pressure bandage: If manual pressure stops the bleeding, replace it with a tight bandage holding the clean dressing firmly in place.

* Tourniquet Protocol (Isolated Settings / Severe Trauma): In rural Morocco, if Civil Protection response times exceed 30 minutes and direct pressure fails to stop catastrophic bleeding (e.g., limb amputation), a tourniquet must be applied.

* Place the tourniquet a few centimeters above the wound (between the wound and the heart), never directly over a joint.

* Tighten until the bleeding stops completely.

* Write down the exact time of application (Format: HH:MM) on the tourniquet or the victim's forehead for the incoming SAMU medical staff. Never loosen a tourniquet once applied.

5. Fractures and Bone Trauma

* Signs and Symptoms: Severe localized pain, visible deformity of the limb, total inability to move, rapid swelling (edema), and bruising.

* Stabilization Protocol:

1. Strict immobilization: First-aid protocols mandate keeping the injured limb completely still. Never attempt to "pop the bone back into place" (reduction), as this can sever blood vessels or nerves.

2. Limb fractures: Immobilize the joints above and below the fracture site using makeshift padded splints (rigid pieces of wood, thick cardboard) secured gently with cloth strips.

3. Open fractures: If the bone pierces the skin, do not try to push it back in. Cover the wound with a sterile dressing or a clean, damp cloth to minimize infection risks before calling 141 or 15.

4. Spinal trauma (Back/Neck): Suspect a spinal cord injury in cases of falls from heights or rural transport accidents (e.g., agricultural machinery, pick-up trucks). Perform manual cervical stabilization by holding the victim's head with both hands in a neutral alignment. Do not move the victim unless there is an immediate threat to life.

6. Cardiac Emergency (Cardiac Arrest)

* Emergency Diagnosis (Rapid Recognition): The victim collapses, is unconscious (does not respond to stimulation or hand squeezes), AND is not breathing (or only exhibits agonal gasps—slow, noisy, labored breathing).

* Adult Cardiopulmonary Resuscitation (CPR) Protocol:

1. Call for help immediately: Dial 141 (SAMU) or 15 (Civil Protection). Clearly state "Cardiac Arrest" to dispatch an advanced life support team.

2. Chest compressions: Lay the victim flat on their back on a firm surface. Place the heel of one hand in the center of the chest (lower half of the breastbone) and interlock your other hand on top.

3. Compress the chest at a rate of 100 to 120 compressions per minute (the tempo of the song Stayin' Alive), to a depth of 5 to 6 cm.

4. Rescue breaths: If trained, alternate 30 chest compressions with 2 rescue breaths. If untrained or unwilling to perform mouth-to-mouth, perform continuous, uninterrupted chest compressions.
5. Defibrillator (AED): If an Automated External Defibrillator is nearby (increasingly available in rural municipal offices, health centers, or transit hubs), turn it on immediately and follow the voice prompts. Do not stop CPR until the device tells you to stand clear for analysis or shock delivery.

7. Snake Envenomation and Spider Bites

* Moroccan Context and Risks: According to the Moroccan Anti-Poison and Pharmacovigilance Center (CAPM), snake envenomation (primarily envenomation by *Cerastes cerastes* or *Daboia mauritanica*) and spider bites (such as *Latrodectus tredecimguttatus*) are frequent seasonal emergencies in rural agricultural regions.

* Snakebite Management Protocol (WHO / Ministry of Health Standards):

1. Reassurance and Immobility: Keep the victim perfectly calm and strictly immobilize the bitten limb using a makeshift sling or loose splint. Movement accelerates the systemic spread of venom through the lymphatic system.
2. Local Wound Care: Clean the bite site gently with clean water or an antiseptic. Remove rings, bracelets, or tight clothing from the affected limb before swelling begins.
3. Strict Prohibitions: Never apply a tourniquet. Never cut the wound open, attempt to suck the venom out (manually or with a pump), or apply ice, heat, or traditional herbal pastes, as these drastically increase local tissue necrosis and infection risks.
4. Emergency Transport: Arrange for immediate medical evacuation to the nearest provincial hospital or rural health center equipped with anti-venom serum. Call 141 or 15 immediately.

* Spider Bite Management Protocol:

1. Clean the Area: Wash the bite thoroughly with soap and water to reduce infection risk.
2. Ice Application: Apply a cold compress or an ice pack wrapped in a clean cloth to the bite site for 10-15 minutes to reduce pain and localized swelling.
3. Monitor for Systemic Symptoms: In rural areas, look out for severe symptoms appearing within hours, such as intense abdominal cramps, muscle rigidity, sweating, or breathing difficulties. Evacuate immediately to a medical facility if these signs manifest.

There are **no private medical clinics or permanent specialist doctors based in Tounfite**. Medicine in this area is entirely public and managed by general practitioners or nurses assigned by the Ministry of Health.

For your medical drone project, your key contacts and the doctors supervising or covering this area in the Midelt province are:

Reference Health Structures (Tounfite & Midelt)

- Tounfite Communal Health Center (CSR2): This is the central public clinic of your deployment zone. State-assigned general practitioners rotate here. There is no direct private phone line for this dispensary; all communications go through the Midelt Health Delegation: +212 5 35 58 20 67.
- Midelt Provincial Hospital: This general hospital supplies all the medications and handles heavy medical emergencies from Tounfite.

Permanent Doctors in the Province (Midelt Reference GPs)

If you need to collaborate with local doctors based in Midelt (the nearest city managing the Tounfite region), here are the practitioners with verified contact lines:

- Dr. Mahmoud MOTIAA (General Practitioner in Midelt)
 - Phone: +212 7 62 91 93 18
 - Utility: Highly recommended locally, ideal for understanding the common medical needs of the region.
- Dr. Mohamed Setihe (Medical Clinic in Midelt)
 - Phone: +212 5 35 58 03 80
 - Utility: General practitioner located on Mohamed V Avenue, experienced with patients coming from rural zones.
- Dr. Mouacha Ahmed (General Practitioner)
 - Phone: +212 6 72 90 47 98
- Dr. Abderrahim Ouaadi (General Practitioner)
 - Phone: +212 6 13 18 79 57

National Medical Emergency Number

To centralize your drone alert system for vital emergencies across the region, the official Moroccan medical emergency number is 141.

Would you like info on how to contact the Midelt Provincial Health Delegation to pitch your drone project for official approval?